New York TLC Project | Statistical Analysis Executive Summary

Overview:

Automatidata

Continuing our project to develop model that predicts fare amount for riders on New York City TLC app, this summary shows results of the work done by our team to perform statistical analysis and hypothesis testing on variables of the dataset.

Details

This stage of the project aimed to find the relation between to important parameters, fare amount and payment method. This could be done using two methods, A/B test from hypothesis testing. This was done using two sample t-test.

Key Insights

- Descriptive statistics were calculated to determine important characteristics of the data such as the center and dispersion.
- About 67% of the data contains rides that were paid using credit cars and 32% by cash.
- Customers who pay in credit card appears to pay a larger fare amount than customers who pay in cash.
- P-value = 6.8 * 10^{-12} , which is less than the significance level. Therefore, we `Reject` the null hypothesis.
- We can conclude that there is a strong relation between fare amount and payment method.
- There might be bias, like the sample was taken contains more representation of credit card payment and also doesn't take into account no-charge or dispute cases.

Next Steps

- We recommend developing a marketing campaign to encourage users to pay using their credit card to gain more revenue.
- In the next step, we’re aiming to perform regression analysis to further investigate the relation between fare amount and payment method.